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Exp No: 14	Packet sniffing using RAW sockets

AIM:

To write a program using raw sockets to capture (sniff) the packets passing through a network interface and to display the Ethernet, IP and TCP / UDP / ICMP header details.

ALGORITHM / PROCEDURE:

1. Create a raw socket with `socket(AF_PACKET, SOCK_RAW, ntohs(3))`. This socket receives every frame (`ETH_P_ALL` = 3) that reaches the network card, with the Ethernet header. Root permission is required.
2. Start an infinite loop and receive a frame with `recvfrom()` (maximum 65535 bytes).
3. Unpack the first 14 bytes as the **Ethernet header**: destination MAC (6 bytes), source MAC (6 bytes) and the type field (2 bytes).
4. If the type is `0x0800` (IPv4), unpack the next 20 bytes as the **IP header** and get the TTL, the protocol number, the source IP and the destination IP. The header length is found from the IHL field.
5. Using the protocol number, unpack the transport header: `1` = ICMP (type, code), `6` = TCP (ports, sequence, acknowledgement), `17` = UDP (ports, length).
6. Display the details of the packet. Repeat until the required number of packets (8) is captured.
7. Generate some traffic in another terminal (ping, UDP echo, browser) and watch the packets being displayed. Close the socket.

Protocol number	Protocol	Header read by the program
1	ICMP	Type, Code
6	TCP	Source port, destination port, sequence no., acknowledgement no.
17	UDP	Source port, destination port, length

DIAGRAM:

Ethernet header 14 bytes	**IP header** 20 bytes	**TCP / UDP / ICMP header** 20 / 8 / 8 bytes	**Data (payload)** remaining bytes
`struct "!6s6sH"`	`struct "!BBHHHBBH4s4s"`	`TCP "!HHLLB" / UDP "!HHH"`	

Packet received by the raw socket (AF_PACKET)

Fig 1: Structure of a captured frame and the struct format used for each header

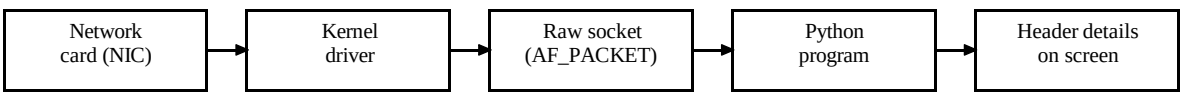


Fig 2: Flow of packet sniffing using a raw socket

PROGRAM:

sniffer.py

```
# Packet sniffer using RAW sockets (Linux) - run as root
import socket, struct

def mac(b):
    return ':'.join('%02x' % x for x in b)

s = socket.socket(socket.AF_PACKET, socket.SOCK_RAW, socket.ntohs(3))
print("Sniffing started... (capturing 8 packets)\n")
count = 0
while count < 8:
    raw, info = s.recvfrom(65535)
    if info[2] == socket.PACKET_OUTGOING:      # skip our own outgoing copy
        continue
    dst, src, proto = struct.unpack("!6s6sH", raw[:14])
    if proto != 0x0800:                    # only IPv4
        continue
    count += 1
    ver_ihl, tos, tlen, ident, frag, ttl, p, chk, sip, dip = struct.unpack("!BBHHHBBH4s4s", raw[14:3
      4])
    ihl = (ver_ihl & 15) * 4
    print(f"Packet {count}  [{info[0]}]")
    print(f"  Ethernet : {mac(src)} -> {mac(dst)}")
    print(f"  IPv4     : {socket.inet_ntoa(sip)} -> {socket.inet_ntoa(dip)}  TTL={ttl}  Length={tlen
      }")
    off = 14 + ihl
    if p == 1:
        t, c = struct.unpack("!BB", raw[off:off+2])
        print(f"  ICMP     : type={t} code={c}")
    elif p == 6:
        sp, dp, seq, ack, offs = struct.unpack("!HHLLB", raw[off:off+13])
        print(f"  TCP      : {sp} -> {dp}  seq={seq}  ack={ack}")
    elif p == 17:
        sp, dp, ln = struct.unpack("!HHH", raw[off:off+6])
        print(f"  UDP      : {sp} -> {dp}  length={ln}")
    else:
        print(f"  Protocol : {p}")
    print()
s.close()
```

OUTPUT:

```
UuUssEeRr @UuUSsEeRr -VirtualBox: ~ (Terminal 1)
UuUssEeRr @UuUSsEeRr -VirtualBox:~$ sudo python3 sniffer.py
Sniffing started... (capturing 8 packets)

Packet 1  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=37
  UDP      : 43740 -> 6000  length=17

Packet 2  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=37
  UDP      : 6000 -> 43740  length=17

Packet 3  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=41
  UDP      : 43740 -> 6000  length=21

Packet 4  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=41
  UDP      : 6000 -> 43740  length=21

Packet 5  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=52
  ICMP     : type=8 code=0

Packet 6  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=52
  ICMP     : type=0 code=0

Packet 7  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=52
  ICMP     : type=8 code=0

Packet 8  [lo]
  Ethernet : 00:00:00:00:00:00 -> 00:00:00:00:00:00
  IPv4     : 127.0.0.1 -> 127.0.0.1  TTL=64  Length=52
  ICMP     : type=0 code=0
```

```
uUssEeRr @uUsSeErR -VirtualBox: ~ (Terminal 2 - traffic generated)
[illegible] [illegible] -VirtualBox:~$ python3 echo_udp_server.py &
[illegible] [illegible] -VirtualBox:~$ python3 echo_udp_client.py
Sent     : Hello UDP
Echoed   : Hello UDP
Sent     : Datagram test
Echoed   : Datagram test
[illegible] [illegible] -VirtualBox:~$ sudo python3 myping.py 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 32 bytes of data.
32 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.015 ms
```

Packet	Protocol	Source	Destination	Details
1, 3	UDP	127.0.0.1 : 43740	127.0.0.1 : 6000	Datagram from the client (17 and 21 byte payload)
2, 4	UDP	127.0.0.1 : 6000	127.0.0.1 : 43740	Echo from the server
5, 7	ICMP	127.0.0.1	127.0.0.1	Type 8 - Echo Request
6, 8	ICMP	127.0.0.1	127.0.0.1	Type 0 - Echo Reply

The capture was taken on the loopback interface `lo`, so the MAC addresses are all zeros. On a real interface (enp0s3) the actual MAC addresses are shown.

RESULT:

Thus a packet sniffer was implemented using raw sockets. The Ethernet, IP and UDP / ICMP header details of the packets passing through the interface were captured and displayed successfully.